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Unpacking the Heat: How Global Energy Imbalance Fuels Wildfires

Abstract

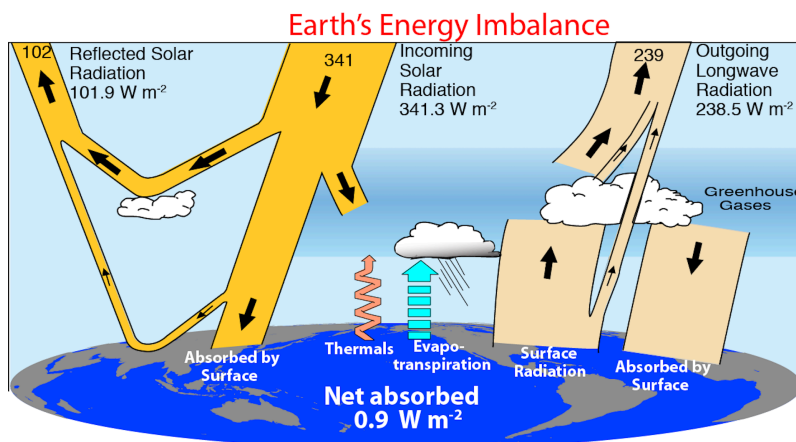
Does climate change cause wildfires? Climate scientist Kevin Trenberth says that is the wrong question. Fires are usually started by people or lightning, but global warming loads the dice by adding extra heat that dries out land and plants. This lesson explains Earth's energy imbalance, where the extra heat goes, and how scientists think about attributing extreme events to climate change.

Introduction

Wildfires burn forests and homes, displace thousands of people and animals, and send smoke hundreds or even thousands of miles. Fighting them has become a year-round task costing billions of dollars. Yet news stories often leave out climate change, partly because it is hard to attribute any single fire to it. After all, there have always been wildfires. Trenberth argues for a different framing, grounded in physics. Global warming does not light the match: that is usually human carelessness or dry lightning. Instead, it raises the risk by setting the stage. To understand how, we need to follow the flow of energy through the climate system.

How Do Scientists Study This?

Trenberth's explanation uses Earth's energy budget. Heat-trapping gases act like a blanket, reducing how much infrared heat escapes to space. Carbon dioxide has increased over 40% since the 1800s, with over half the increase since 1985. Scientists estimate the resulting energy imbalance at about 1 watt per square meter, or about 500 terawatts globally, by tracking where heat accumulates: mostly the ocean, measured down to 2,000 meters, plus melting ice. On land, water acts as the planet's air conditioner: heat goes into evaporation when soil is moist. Trenberth then calculated how much energy builds up during a dry spell when there is no water to absorb it.



FROM THE ARTICLE

Earth's energy flows: incoming solar, reflected, and outgoing radiation, with a net absorbed imbalance at the surface. Trenberth et al. (2009).

How extra heat sets the stage for fire

1 Greenhouse gases

CO₂ up over 40% since the 1800s

2 Energy imbalance

About 1 W per square meter

3 Dry spell

No water to absorb the heat

4 Fire risk

Plants dry out; heat builds

Study the Process

Measure the imbalance

Scientists estimate Earth absorbs about 1 watt per square meter more than it releases.

Track the heat

Ocean heat content, melting ice, and land temperatures show where the energy goes.

Follow the water

When land is dry, extra heat builds up, drying plants and raising temperatures.

What Did They Discover?

Earth's energy imbalance of about 500 terawatts is small compared with the natural energy flow of 240 watts per square meter, but huge compared with human activities: U.S. electricity generation averaged just 0.46 terawatts. Over 90% of the extra heat ends up in the ocean, and sea level is rising over 3 millimeters per year. On land, if the imbalance accumulates over a month without rain, it equals running a small microwave oven at full power over every square foot for six minutes. Because soil moisture and plant roots delay the effect, it typically takes over two months of dry weather to fully set the stage for wildfires, as seen in the western U.S. and Mediterranean regions.

Earth's energy imbalance

500 TW

extra energy absorbed by Earth globally

0.46 TW

average U.S. electricity generation, for comparison

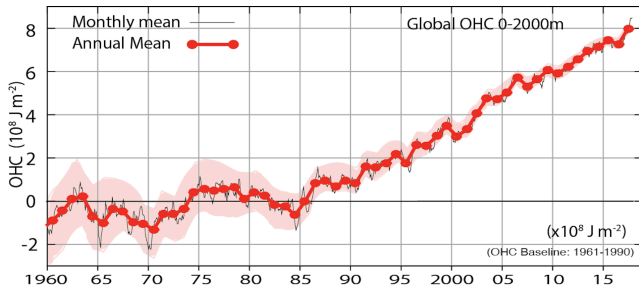
90%+

of the extra heat ends up in the ocean

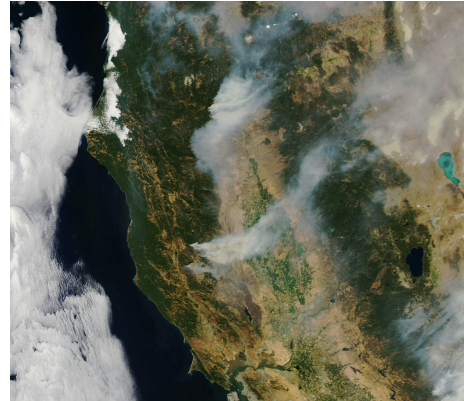
Scientists no longer say events are due to global warming; instead, they say extreme events would very likely not have been so severe without it.

Look at the Evidence

FROM THE ARTICLE



Global ocean heat content for the top 2,000 meters, with uncertainty shown in pink. Science Advances, CC BY-NC.



A satellite image of the Carr fire in California, fueled by drought and dead vegetation. Image: NASA.

Takeaways

Warming raises risk

Climate change does not start fires, but extra heat dries land and makes fires more likely.

Water cools the land

Without rain, excess heat accumulates, leading to heat waves and fire danger.

Reframe attribution

All weather now occurs in a warmer, moister environment shaped by climate change.

Key Terms

Energy imbalance

The difference between the energy Earth absorbs from the Sun and the energy it radiates back to space.

Watt per square meter

A unit of energy flow over an area, used to describe how much energy reaches or leaves Earth's surface.

Ocean heat content

The total heat stored in the ocean, which has been rising as the planet warms.

Attribution science

Research that determines how much climate change influenced a particular extreme event.

Anticyclone

A high-pressure weather system that can stall and bring long stretches of dry, hot weather.

Words to Know · Unpacking the Heat

Energy imbalance

Watt per square meter

Ocean heat content

Attribution science

Anticyclone

44 | ACTIVITY

Activity

Microwave Math

Students recreate Trenberth's calculation to understand how a small energy imbalance adds up during a dry spell.

Materials: Calculator; Worksheet with the lesson's numbers; Graph paper

1. Start with an imbalance of 1 watt per square meter. Calculate the energy over 720 hours (30 days).
2. Compare with a small microwave at about 720 watts: how long would it run on one square meter to match?
3. Graph how accumulated energy grows day by day during a 60-day dry spell.
4. Explain why moist soil and plant roots delay the effect, and why rain resets it.
5. Write a short paragraph answering: does climate change cause wildfires?

References

Trenberth, K. (2018). Climate change and wildfires: how do we know if there is a link? The Conversation. <https://doi.org/10.64628/AAI.h3xhvwcfm>